

Spatially Resolved Transcriptomics of the Small Intestinal Villus and Its Microbiome Sensitivity

Replication and Extension of Mayassi et al. 2024

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Modeling of Complex Biological Systems

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Abstract

The small intestinal epithelium is organized along two spatial axes: a proximal-to-distal axis spanning duodenum, jejunum, and ileum, and a vertical crypt-to-villus axis along which stem cells differentiate into mature absorptive enterocytes. While Mayassi et al. 2024 comprehensively characterized regional transcriptional identity across the gut using Visium spatial transcriptomics, their analysis focused primarily on the colon, leaving the small intestinal villus layer architecture largely uncharacterized. Here I replicate their spatial framework and extend it by asking two questions: whether the crypt-to-villus transcriptional gradient is conserved across SI segments, and whether the microbiome differentially affects gene expression along this axis. I find that a large conserved core program exists across all four SI segments — 646 crypt markers and 2,381 villus tip markers shared across duodenum, jejunum1, jejunum2, and ileum — overlaid with region-specific gene expression that increases toward the villus tip. GO enrichment confirms the biological validity of these layers: conserved crypt markers are enriched for cell cycle and DNA replication terms, while conserved tip markers reflect the secretory pathway and vesicle transport. Comparing germ-free and specific-pathogen-free mice, I find that the microbiome asymmetrically affects the crypt-to-villus axis: the bottom villous SI is most transcriptionally promoted by microbiome presence, while the villus tip shows unexpected upregulation of mRNA processing, translation, and protein catabolism in germ-free mice, suggesting the microbiome induces transcriptional restraint at the most exposed epithelial surface.

1 Introduction

The mammalian small intestine is responsible for the absorption of virtually all dietary nutrients, a function carried out by a specialized epithelium organized into villi. Each villus is a dynamic structure maintained by a continuous process of cell renewal: intestinal stem cells at the base of crypts divide and give rise to transit-amplifying progenitors, which migrate upward along the villus, differentiating into mature absorptive enterocytes as they go, and eventually shedding from the villus tip into the gut lumen [1]. This crypt-to-villus axis represents one of the most visible differentiation gradients in mammalian biology, with cells transitioning from rapidly dividing stem cells to post-mitotic, functionally specialized absorptive cells over a period of just three to five days [2].

The three major segments of the small intestine — duodenum, jejunum, and ileum — each perform distinct physiological roles. The duodenum receives preprocessed food from the stomach and neutralizes gastric acid, while absorbing iron, calcium and fat-soluble vitamins (A, D, E, K). The jejunum is the primary site of monosaccharides, amino acid, and lipid absorption. The ileum specializes in bile acid reabsorption, water-soluble vitamins (B12, C) uptake, and contains the highest density of immune cells [3]. These regional differences are reflected in distinct transcriptional programs, though the extent to which the crypt-to-villus axis itself differs across segments has not been well characterized.

Mayassi et al. 2024 performed a landmark spatial transcriptomics study of the mouse gut using Visium technology, generating a comprehensive atlas of 120,000 tissue spots spanning the entire gastrointestinal tract [4]. Their key finding was that regional transcriptional identity is remarkably robust to microbiome perturbation — germ-free and conventionally housed mice maintain similar spatial gene expression patterns across the gut. They further identified a microbiota-driven transcriptional region in the mid-colon. However, their analysis focused predominantly on colonic regionalization, since that is where they found the most distinctive patterns, and the spatial architecture of the small intestinal villus — particularly the crypt-to-villus axis — was not deeply characterized.

In this work, I replicate parts of the spatial framework established by Mayassi et al. and extend the analysis to the small intestinal villus layer axis. I address two questions: (1) Is the crypt-to-villus transcriptional gradient conserved across SI segments, or does each segment have a unique villus program? (2) Does microbiome presence differentially affect gene expression along the crypt-to-villus axis? I hypothesized that a conserved core villus program exists across segments, and that the villus tip — directly exposed to luminal bacteria — would be more microbiome-sensitive than the protected crypt.

2 Methods and Materials

2.1 Data

Visium spatial transcriptomics data were obtained from the Broad Single Cell Portal (accession SCP2762), corresponding to the microbiome experiment described in Mayassi et al. 2024. The dataset comprises 120,021 Visium spots from mouse small intestine and colon, spanning four small intestinal regions (Duodenum, Jejunum1, Jejunum2, Ileum) and colon, across three microbiome conditions: specific pathogen-free (SPF), germ-free (GF), and fecal microbiota transplant (FMT). Each spot captures RNA from approximately 5-15 cells at a 55 μm diameter resolution. Spot-level metadata including tissue region, villus layer annotation (ylayer), and microbiome condition were provided by the authors. The ylayer annotation assigns each spot to one of four categories: Crypt SI, Bottom villous SI, Top villous SI, or Muscle SI.

2.2 Preprocessing

Raw count matrices were loaded using `scipy.io` and assembled into `AnnData` objects using `Scanpy` v1.12.1 [5]. Quality control metrics were calculated using `sc.pp.calculate_qc_metrics`, flagging mitochondrial genes by the `mt-` prefix. Spots with fewer than 500 counts, fewer than 200 detected genes, or more than 20% mitochondrial counts were removed as likely damaged tissue (standard thresholds), resulting in 119,892 spots retained for analysis (the data was already preprocessed by the authors). Highly variable genes were selected on raw counts using the `seurat_v3` method ($n=3,000$). Raw counts were stored prior to normalization. Spots were normalized to 10,000 counts per spot and $\log_1p$ (to avoid $\log(0)$) transformed. Principal component analysis was performed on the 3,000 highly variable genes (`zero_center=False` to preserve sparsity). A k-nearest neighbor graph was computed ($k=15$, $n_pcs=30$) and UMAP embedding was computed for visualization.

2.3 Leiden Clustering

Leiden unsupervised clustering was performed on the data using several resolution parameters (0.02, 0.05, 0.1, 0.25, and 0.5) to check whether independent clustering would recapture the known regional and villus layer structure annotated by the authors. Clustering at resolution = 0.5 yielded 27 clusters that are largely region-pure (Figure 1), but each region was split into multiple sub-clusters corresponding to the “petal” substructure of the UMAP, not to the known villus layers. At the lowest resolution (0.02), the clustering produced 4 clusters — the same number as the annotated regions — but these clusters were not pure, with the two jejunum regions mixing the most. This suggests that both the proximal-distal axis and, even more so, the crypt-to-villus axis form a continuous gradient, and that spatially adjacent regions cannot be easily separated by unsupervised clustering. For this reason, I used the pre-existing expert annotations (tissue region and ylayer) directly for the differential expression analysis.

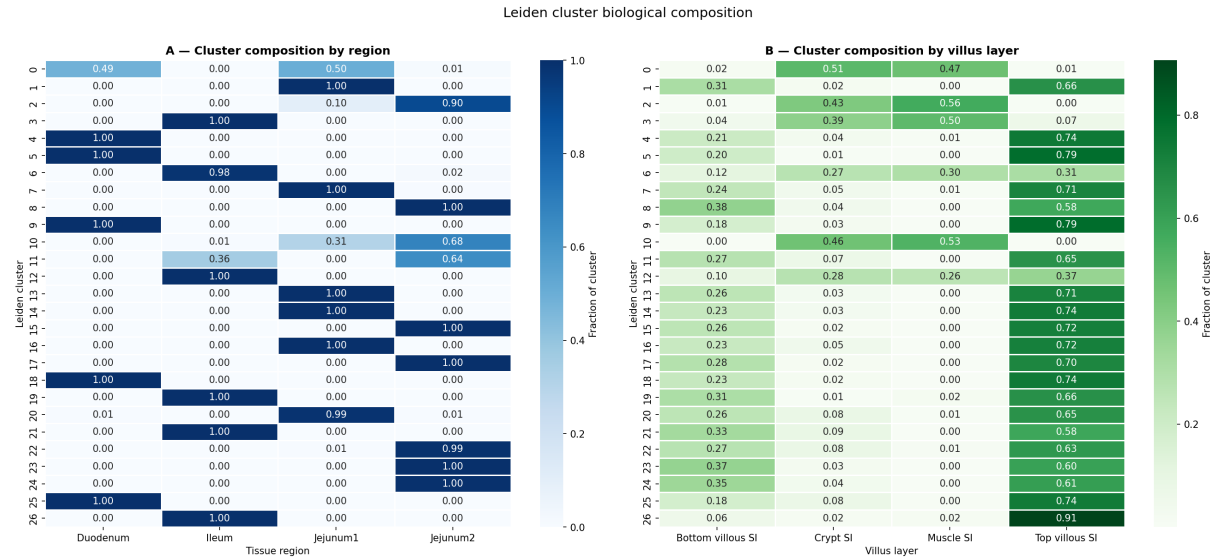


Figure 1: Composition of Leiden clusters (resolution=0.5) with respect to tissue region (A) and villus layer (B). Each row shows, for one Leiden cluster, the fraction of its spots belonging to each category. Most clusters are dominated by a single region (Panel A), with the two jejunum regions showing the most mixing, while no cluster is dominated by a single villus layer (Panel B).

2.4 Differential Expression Analysis

Differential expression analysis was performed using the Wilcoxon rank-sum test implemented in `Scanpy` (`sc.tl.rank_genes_groups`). This non-parametric test was chosen because gene

expression data in spatial transcriptomics is highly skewed and that violates the normality assumption of parametric tests. For regional marker identification, each SI region was compared against all other regions (one-vs-rest). For villus layer marker identification, analysis was restricted to epithelial spots (Crypt SI, Bottom villous SI, Top villous SI). Muscle SI was excluded since it does not represent an epithelial layer and so would only be confounding for the analysis. Significant DEGs were defined by adjusted p-value < 0.05 and $|\log_2 FC| > 0.5$. For the microbiome comparison (SPF vs GF), the larger condition was downsampled to match the smaller condition within each region $\times$ layer combination prior to analysis (random seed = 42) to ensure equal statistical power. A minimum log2 fold change threshold of 0.25 was applied for directional analysis. The lower log2 fold change threshold for the microbiome comparison was chosen to capture the expected subtler differences.

2.5 Conservation Analysis

To assess conservation of villus layer markers across SI segments, DEG analysis was performed independently within each of the four SI regions for each villus layer. Genes identified as significant markers in all four regions were classified as conserved core markers. Genes significant in only one region were classified as region-specific markers.

2.6 Gene Ontology Enrichment

GO enrichment analysis was performed using gseapy v1.2.1 [6] via the Enrichr API, querying the GO Biological Process 2023 database. Mouse gene symbols were used directly. Significant GO terms were defined by adjusted p-value < 0.05 . Mitochondrial genes were excluded from visualization of marker gene expression heatmaps, as they reflect general cellular energy demand rather than layer-specific functional identity.

2.7 Reproducibility

All analyses were performed in Python 3.11 within a conda environment. The complete workflow is defined in a Snakefile and executed via Snakemake v9.22.0 using papermill v2.7.0 for notebook execution. Code and notebooks are available at <https://github.com/FairOn/gut-transcriptomics-project>.

3 Results

3.1 Replication of Spatial Framework

I successfully replicated the spatial transcriptomics framework established by Mayassi et al. 2024. The unrolled gut visualization (Figure 2) reproduces the tissue architecture described in the original paper, with the proximal-to-distal gut axis represented on the X axis and the crypt-to-villus axis on the Y axis. The villus layer annotations (Crypt SI, Bottom villous SI, Top villous SI, Muscle SI) are clearly visible as distinct horizontal bands within each regional section.

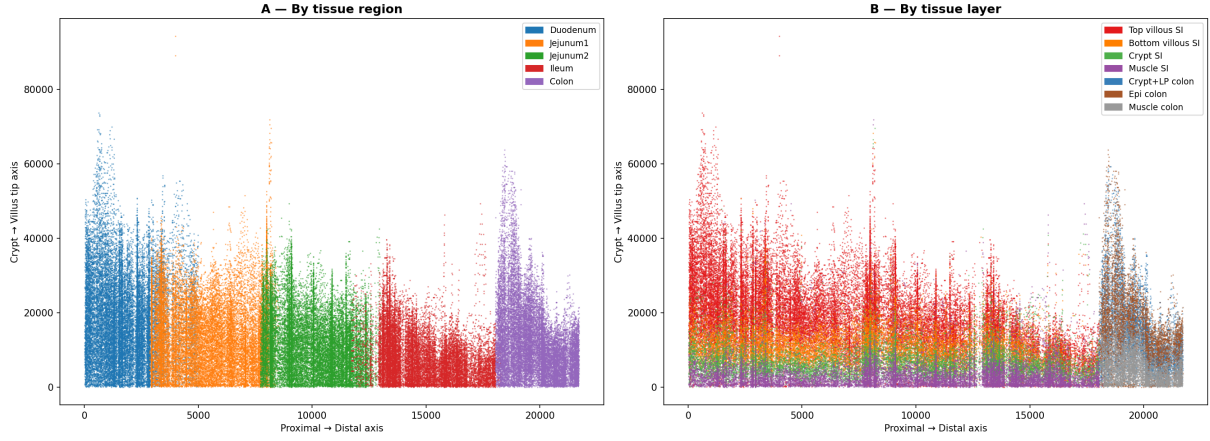


Figure 2: Unrolled gut spatial transcriptomics overview. Left: spots colored by tissue region. Right: spots colored by villus layer annotation. The X axis represents the proximal-to-distal gut axis; the Y axis represents the crypt-to-villus tip axis.

UMAP embedding of all 119,892 spots (Figure 3) shows clear separation between gut regions, with small intestine and colon forming distinct clusters. SPF and GF spots intermix throughout the UMAP, confirming the paper's core finding that regional transcriptional identity is robust to microbiome presence.

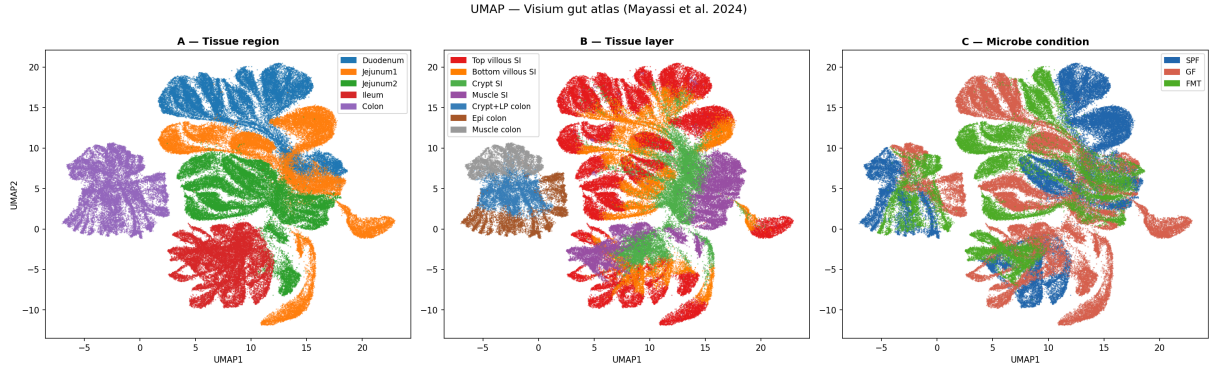


Figure 3: UMAP of all Visium spots colored by tissue region (A), villus layer (B), and microbiome condition (C). SPF and GF spots intermix throughout, confirming transcriptional robustness to microbiome perturbation.

3.2 Small Intestinal Regional Identity

Restricting analysis to small intestine spots and recomputing the UMAP provides a more detailed view of SI-specific transcriptional structure (Figure 4). All four SI regions form distinct clusters with an internal "flower" structure reflecting the villus layer organization within each segment.

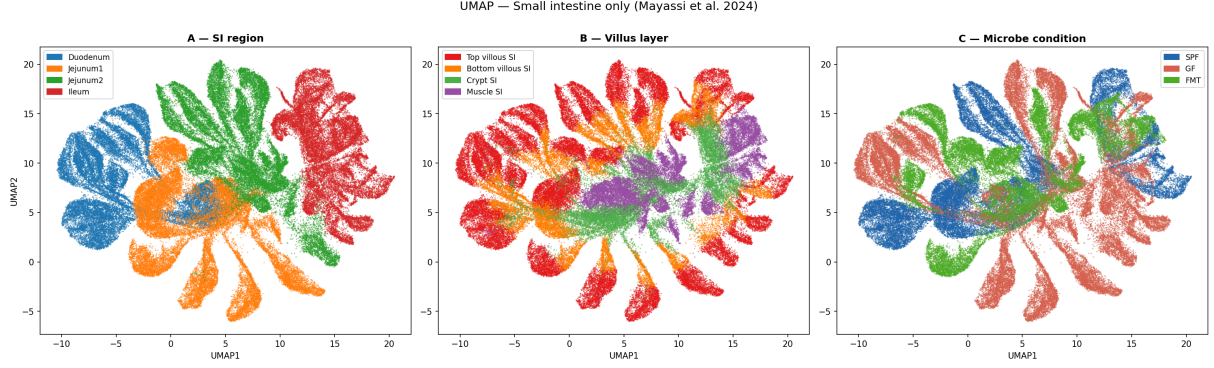


Figure 4: UMAP of small intestine spots only, colored by region (A), villus layer (B), and microbiome condition (C). Each region forms a distinct cluster with internal villus layer structure.

Differential expression analysis identified 6,406 significant regional marker genes across all four SI segments. The ileum showed the most unique transcriptional identity (2,801 DEGs), followed by duodenum (1,929), jejunum1 (1,075), and jejunum2 (601). Top marker genes for each region are biologically validated: duodenum markers include *Reg1* and *S100g*, jejunum markers include *Fabp1* and *Apoa4* (fatty acid and lipoprotein absorption), and ileum markers include *Fabp6* (bile acid binding) and *Tm6gd1* (Figure 5).

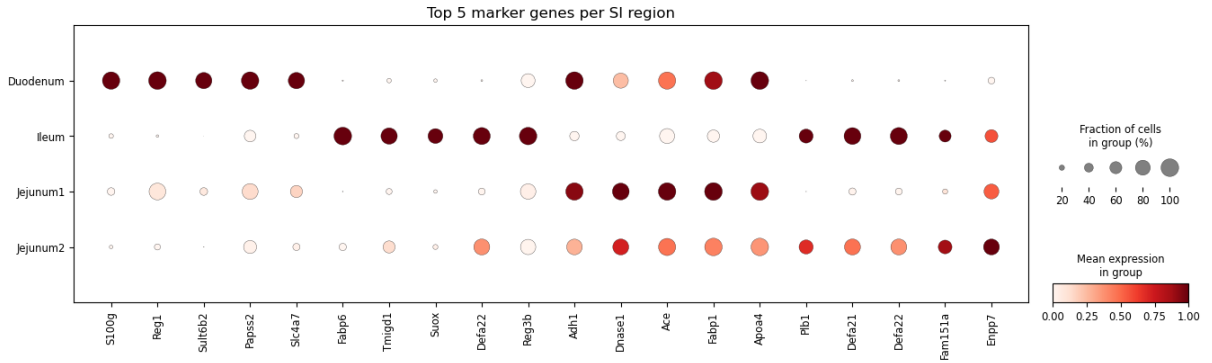


Figure 5: Dotplot of top 5 marker genes per SI region. Dot size indicates the fraction of spots expressing the gene; color indicates mean expression level (scaled 0-1 per gene).

3.3 Conservation of the Crypt-to-Villus Transcriptional Gradient

To characterize the crypt-to-villus axis, I performed differential expression analysis between Crypt SI, Bottom villous SI, and Top villous SI spots, first across all SI regions combined and then independently within each region. Across all regions, 8,870 significant layer DEGs were identified, with Top villous SI showing the most (4,313), followed by Bottom villous SI (3,245) and Crypt SI (1,312), which is not surprising given the increasing functional specialization present toward the villus tip.

Crypt SI markers are dominated by proliferation and cell cycle genes, including *Mki67* (canonical proliferation marker), *Olfm4* (intestinal stem cell marker), and *Stmn1* (expressed in rapidly dividing progenitors), validating the biological accuracy of the layer annotations (Figure 6).

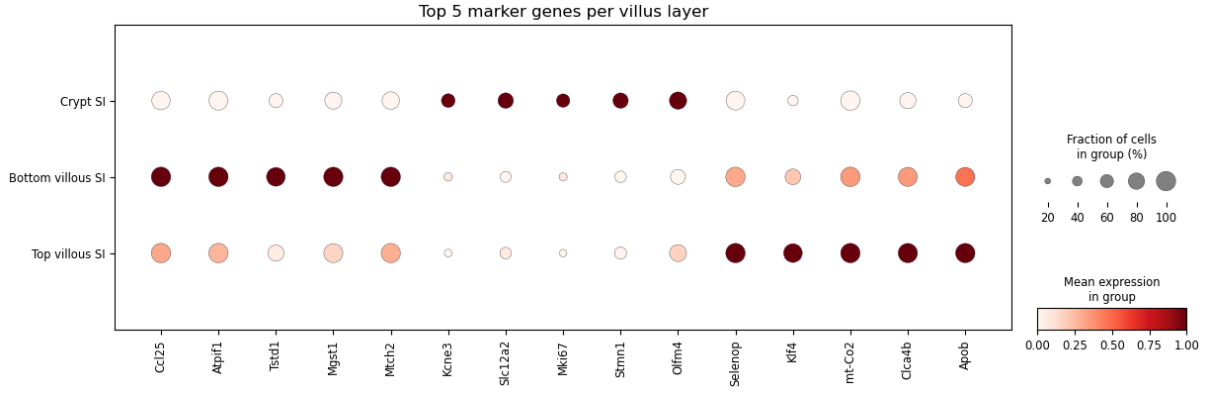


Figure 6: Dotplot of top 5 marker genes per villus layer. Layers are ordered from crypt base (bottom) to villus tip (top), reflecting the direction of cell migration. Key markers include *Mki67* and *Olm4* (Crypt SI) and *Apob* and *Klf4* (Top villous SI).

Conservation analysis revealed that a large conserved core program is shared across all four SI segments for each villus layer (Figure 7): 646 conserved crypt markers, 2,224 conserved bottom villous markers, and 2,381 conserved top villous markers. The villus tip shows both the largest conserved core and the largest region-specific component, with the ileum contributing the most unique tip markers (975 region-specific genes). This is most likely because the cells at the villus tip are the most specialized, and therefore they have both the highest number of markers overall and the highest number of region-specific markers.

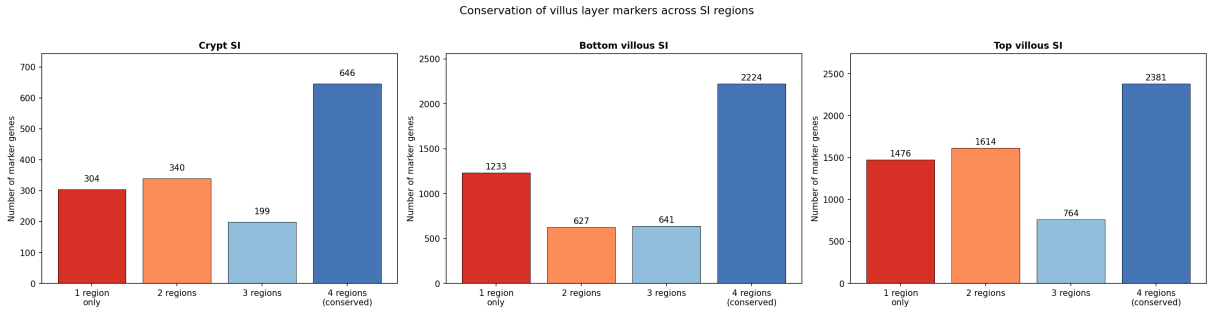


Figure 7: Conservation of villus layer markers across SI regions. For each layer, marker genes are classified by how many regions they appear in (1 = region-specific, 4 = conserved across all segments). The blue bar represents the conserved core shared by all four SI regions.

Expression of the top conserved villus tip markers (excluding mitochondrial genes) across all region-layer combinations confirms a gradient of increasing expression from crypt to villus tip (Figure 8). Key conserved tip markers include *Apob* (apolipoprotein B, required for chylomicron assembly and fat absorption), *Klf4* (transcription factor marking terminally differentiated enterocytes), *Slc6a19* (neutral amino acid transporter), and MHC class I molecules *H2-K1* and *H2-D1*.

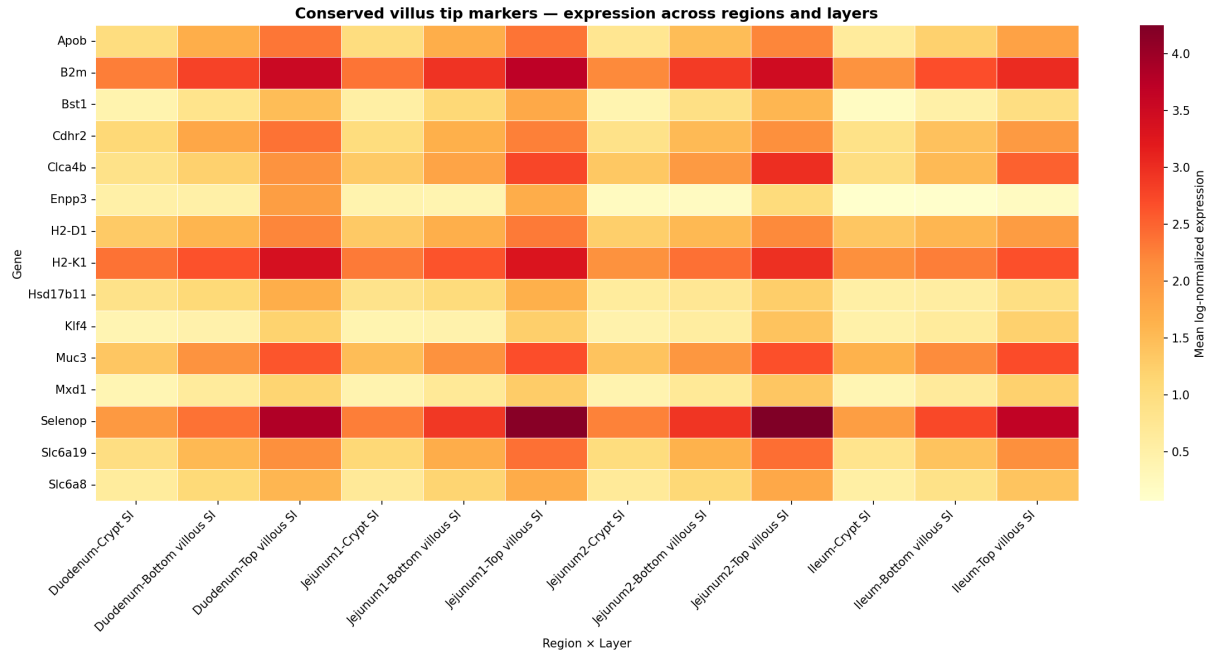


Figure 8: Mean expression of top conserved villus tip markers across all region-layer combinations. Columns are ordered Crypt → Bottom villous → Top villous within each region, reflecting the biological gradient. Mitochondrial genes excluded for clarity.

3.4 Gene Ontology Enrichment of Villus Layer Markers

GO enrichment of conserved layer markers confirms the biological identity of each layer (Figure 9). Conserved Crypt SI markers are highly enriched for cell cycle and DNA replication terms (Mitotic Sister Chromatid Segregation, DNA Metabolic Process, Positive Regulation of Cell Cycle Process), consistent with the crypt's role as the proliferative compartment. Conserved Top villous SI markers are enriched for vesicle transport and secretory pathway terms (Golgi Vesicle Transport, ER-to-Golgi Transport, Vesicle-Mediated Transport), reflecting the continuous secretion of chylomicrons, mucus, and antimicrobial peptides by mature enterocytes at the villus tip.

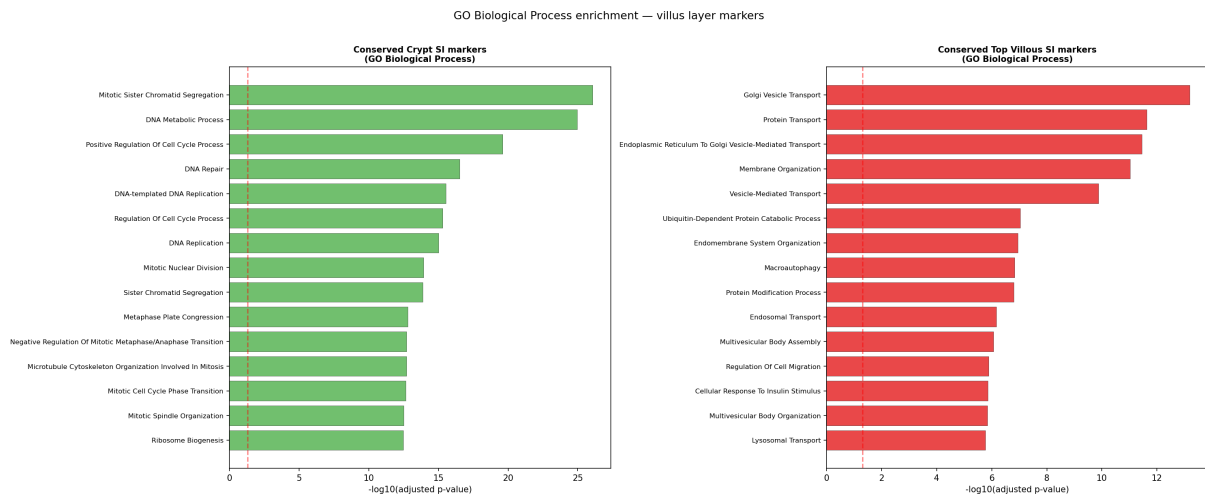


Figure 9: GO Biological Process enrichment for conserved Crypt SI markers (green) and conserved Top villous SI markers (red). Bars show $-\log_{10}(\text{adjusted p-value})$. Red dashed line marks the $p=0.05$ significance threshold.

3.5 Microbiome Differentially Affects the Crypt-to-Villus Axis

To test whether the microbiome affects gene expression differently along the crypt-to-villus axis, I compared SPF and GF spots within each of the 12 region-layer combinations, downsampling to equal spot counts per condition to ensure fair comparison. The results reveal a striking asymmetry (Figure 10).

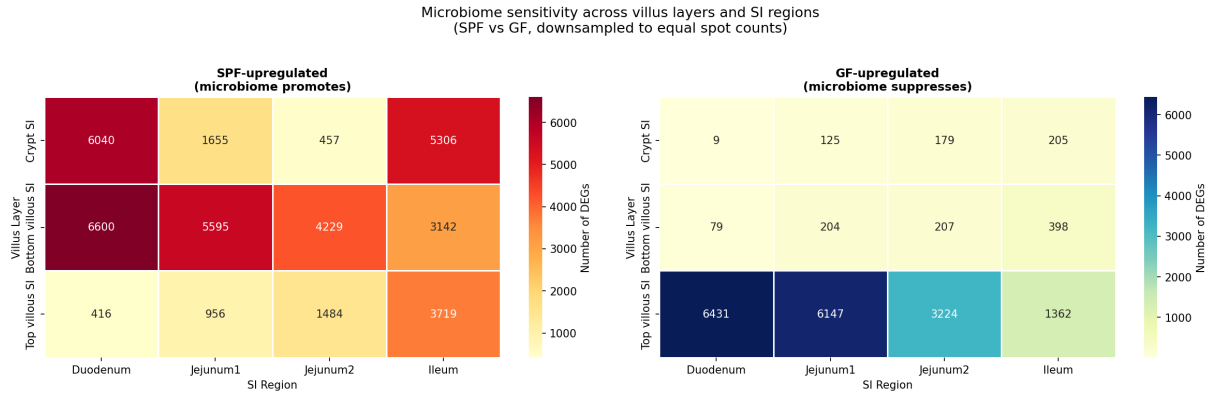


Figure 10: Number of SPF vs GF differentially expressed genes per region-layer combination. Left: genes upregulated in SPF (promoted by microbiome). Right: genes upregulated in GF (suppressed by microbiome). Both conditions were downsampled to equal spot counts prior to analysis.

The Bottom villous SI is the most microbiome-promoted layer across three out of four of the SI regions (3,142-6,600 SPF-upregulated DEGs), suggesting the microbiome primarily influences the enterocyte maturation zone rather than stem cell identity or fully differentiated cells.

Contrary to my initial hypothesis, the Top villous SI shows the largest GF-upregulation rather than SPF-upregulation — thousands of genes are upregulated at the villus tip specifically in germ-free mice (6,431 in Duodenum, 6,147 in Jejunum1, 3,224 in Jejunum2, 1,362 in Ileum). GO enrichment of these GF-upregulated top villous genes reveals enrichment for mRNA Processing, Gene Expression, RNA Splicing, Translation, and Protein Catabolism (Figure 11).

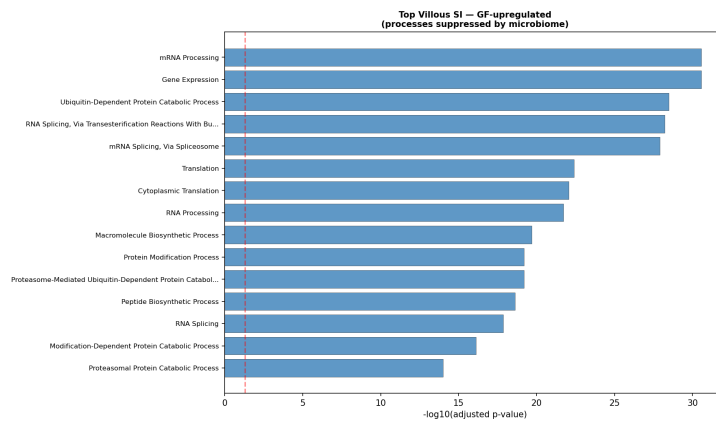


Figure 11: GO Biological Process enrichment for genes upregulated at the villus tip in germ-free mice — processes suppressed by the microbiome at the villus tip.

These results indicate that the microbiome suppresses gene expression and protein synthesis machinery at the villus tip in normal mice. In germ-free mice, this suppression is absent, leading to upregulation of transcriptional and translational processes at the most exposed epithelial surface.

4 Discussion

In this study I characterized the transcriptional architecture of the small intestinal villus using Visium spatial transcriptomics data from Mayassi et al. 2024, extending their analysis from a colon-focused perspective to an examination of the crypt-to-villus axis across all four SI segments.

My first main finding is that the crypt-to-villus transcriptional gradient is organized as a conserved core program shared across duodenum, jejunum, and ileum, overlaid with region-specific gene expression that increases toward the villus tip. The conserved crypt program reflects universal stem cell and proliferation machinery — consistent with the known biology of intestinal stem cells [7]. The conserved tip program reflects the secretory pathway and vesicle transport, consistent with the known function of mature enterocytes in secreting chylomicrons via the Golgi apparatus [8]. The increasing region-specificity toward the villus tip takes place, probably, due to the segment-specific functional specializations of mature enterocytes — duodenal enterocytes absorb iron and calcium, jejunal enterocytes absorb fatty acids and amino acids via distinct transporter sets, and ileal enterocytes reabsorb bile acids via specific transporters.

My second main finding concerns the microbiome’s differential effect along the crypt-to-villus axis. I initially hypothesized that the villus tip, being directly exposed to the gut lumen, would show the greatest microbiome sensitivity. This hypothesis was not fully confirmed. Instead, I found that the bottom villous SI is the most microbiome-promoted layer, and — more surprisingly — that the villus tip is transcriptionally hyperactive in germ-free mice, with upregulation of mRNA processing, translation, and protein catabolism. One possible interpretation is that this represents microbiome-induced transcriptional suppression at the villus tip, potentially preventing excessive inflammatory responses to luminal bacteria. However, alternative explanations — such as compensatory upregulation in the absence of microbial stimulation, or morphological differences in GF villus structure — cannot be excluded with the current data. This is consistent with the known role of commensal bacteria in maintaining intestinal immune homeostasis [9], and with evidence from gnotobiotic mouse models that microbiota presence shapes small intestinal villus structure and development [10].

4.1 Limitations

Several limitations should be noted. First, Visium technology captures RNA from multiple cells per spot (5-15 cells at 55 μm resolution), meaning my layer-level analyses reflect averaged expression rather than true single-cell resolution. Future analyses using single-cell RNA sequencing or higher-resolution spatial methods (such as Xenium, which is also available in the Mayassi et al. dataset) could provide finer characterization of the crypt-to-villus gradient. Second, all data are from mouse tissue, and while mouse and human intestinal biology are broadly conserved, direct translation to human disease requires validation in human samples. Third, the microbiome comparison is limited to SPF vs GF; the FMT condition, which would allow assessment of microbiome rescue effects, was not analyzed due to time constraints. Fourth, the number of DEGs detected is influenced by spot count even after downsampling, as larger absolute numbers of spots per condition increase statistical power for detecting small effect sizes. Finally, the villus tip markers identified here could be contextualized against published transcriptomic datasets from intestinal disease models — such as celiac disease, where villous atrophy selectively destroys the tip compartment — to directly quantify which conserved tip programs are disrupted during pathology.

4.2 Conclusions

I have shown that the small intestinal villus is organized as a conserved transcriptional gradient shared across all SI segments, with region-specific gene expression increasing toward the absorptive surface. The microbiome asymmetrically affects this gradient, primarily promoting expression in

the transitional bottom villous zone and suppressing gene expression machinery at the villus tip. These findings provide a spatially resolved molecular baseline for the small intestinal epithelium that could inform future studies of villous atrophy in celiac disease and other enteropathies.

References

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Note on the Use of Large Language Models

This project was completed with the assistance of Claude (Anthropic), a large language model. It was used to check the correctness of the code, provide insights on the technical side of the code, and to help with the generation of snakemake file and template for this report.